MACRONEWS PORTFOLIO LAB: A NEWS- AND MOMENTUM-DRIVEN PORTFOLIO BACKTESTING SIMULATOR FOR THE U.S. MARKET

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1 Group Members

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2 What is your project about?

This project builds *MacroNews Portfolio Lab*, a web-based simulator that backtests portfolio allocation strategies on U.S. sector ETFs using financial news and price signals. We collect Federal Reserve communications, macroeconomic and sector news, and fine-tune FinBERT to classify each text by macro topic (monetary policy, inflation, labor market, growth, energy, global economy) and to score its sentiment and hawkish/dovish stance. These scores are aggregated into weekly news indices that generate return views, while price momentum and short-term reversal generate a second, technical set of views. The Black–Litterman model combines both sets of views with market-implied returns, and a long-only mean-variance optimizer rebalances the portfolio every week. Users can select assets, periods, strategies and parameters, run backtests on historical data and compare the results; the platform does not perform any real trading.

3 Why this project?

Federal Reserve decisions and macroeconomic news drive large shifts across U.S. sectors, yet translating a continuous stream of text into portfolio decisions remains difficult for investors. At the same time, classical mean-variance optimization is highly sensitive to estimated expected returns and often produces extreme, unstable portfolios. The Black–Litterman model addresses this by blending a neutral market prior with multiple sources of views in a Bayesian way, each weighted by its own confidence, which makes it a natural

framework for combining text signals with price-based signals. Our guiding question is whether news adds information beyond what prices already reflect, i.e. whether news-based views improve risk-adjusted returns or reduce drawdowns compared with momentum-only portfolios. Rich U.S. data also allow a reliable evaluation across NLP, deep learning, data mining and quantitative finance.

4 How will you do it?

The system consists of an offline pipeline and an online simulator. The offline pipeline, run once, collects texts and prices, fine-tunes FinBERT on public labelled datasets plus a small manually labelled macro-topic set (with inter-annotator agreement measured by Cohen's kappa), scores all texts and stores weekly news indices in a MySQL database. The simulator runs a walk-forward backtest over a user-selected period: each week it estimates a covariance matrix, fits a Ridge regression on past data only to map news indices into news views, computes technical views from momentum and reversal, combines both through Black–Litterman with separate confidence levels and solves a long-only mean-variance problem, so no future information is used. We evaluate (i) fine-tuned FinBERT against Loughran–McDonald lexicon and TF-IDF baselines using macro-F1; (ii) an ablation of Black–Litterman with no views, technical views, news views and both, against equal-weight, minimum-variance, inverse-volatility, Markowitz and SPY buy-and-hold, plus a placebo with shuffled news, using the Sharpe ratio, maximum drawdown, CVaR and turnover with bootstrap tests; and (iii) robustness during stress periods such as 2020 and 2022. Data are used under their terms of use, and the simulator is for educational purposes only, not investment advice.

Component	Method	Tools
Data	FOMC statements and minutes, public financial news datasets, FRED macro series; daily prices of 11 SPDR sector ETFs, Treasury and gold ETFs, SPY (2010–2023)	yfinance, fredapi, pandas, MySQL
NLP	FinBERT fine-tuned for topic, sentiment and hawkish/dovish stance (Financial PhraseBank, labelled FOMC data, own labels); lexicon and TF-IDF baselines	PyTorch, Transformers, Label Studio
Portfolio	Weekly news indices; Ridge-based news views; momentum and reversal views; Ledoit–Wolf covariance; Black–Litterman; long-only mean-variance	NumPy, scikit-learn, cvxpy
Evaluation	Walk-forward weekly backtest with costs; ablation and placebo tests; Sharpe, max drawdown, CVaR, turnover; bootstrap tests	pandas, NumPy
Platform	Backtest simulator with strategy and run comparison	FastAPI, React, Docker

Timeline	Milestone
Weeks 3–8	Project description (Oct 9); data pipeline, labelling, baselines; progress report (Nov 13)
Weeks 9–13	FinBERT fine-tuning, news views, Black–Litterman integration, full backtests
Weeks 14–15	Simulator completion, evaluation and expert feedback
Week 16 / 17	Final presentation (Jan 8) / Final report (Jan 15)

5 What do you expect to produce?

- A dataset of scored financial texts and weekly news indices, and a fine-tuned FinBERT model
- A Black–Litterman portfolio engine with news and technical views, and a walk-forward backtester
- A web-based backtest simulator (FastAPI + React) packaged with Docker
- A final report and presentation with the evaluation results